

Phase 0 – Project Foundation (Day 1)

Goal: Set up a production-ready codebase.

Deliverables

- React + Vite frontend
- FastAPI backend
- Folder structure
- Tailwind CSS + shadcn/ui
- Environment variable management
- API communication between frontend and backend
- Docker support (optional)
- Logging
- Error handling
- Git repository

Outcome

Frontend ↔ Backend

Nothing AI-related yet.

Phase 1 – Authentication

Goal: Users can securely log in with Google.

Features

- Google OAuth 2.0
- Sign in
- Sign out
- Store user profile
- Refresh tokens
- Secure backend session
- Protect routes

Outcome

User

↓

Google Login



Dashboard

Phase 2 – Google API Integration

Goal: Verify backend access to the user's Google account.

Features

- Connect Google Forms API
- Connect Google Drive API
- Connect Google Sheets API
- Test permissions
- List existing forms
- Create a blank form

Example

Click Button



Blank Google Form Created



Display Edit Link

At this stage, there is no AI.

Phase 3 – Form Builder Engine (No AI)

Goal: Build the core engine that creates forms from structured JSON.

Example input:

{

```
"title": "Student Registration",  
  
"questions": [  
  
  {  
  
    "title": "Full Name",  
  
    "type": "SHORT_TEXT"  
  
  }  
  
]  
}
```

The backend should generate the Google Form from this JSON.

Features

- Create form
- Add questions
- Multiple choice
- Dropdown
- Checkboxes
- Paragraph
- Date
- Time
- Rating
- File upload (if supported by API/user permissions)
- Required fields
- Sections

Outcome

A working form generator independent of AI.

Phase 4 – AI Prompt → JSON

Now introduce the LLM.

Input:

Create a registration form for Annual Gathering.

Output:

```
{
```

```
"title": "...",  
"description": "...",  
"questions": [...]  
}
```

The LLM does **not** call Google APIs yet.

Features

- Prompt understanding
- Question generation
- Question types
- Required fields
- Validation

Phase 5 – AI + Form Engine

Connect the LLM output to the form generator.

Prompt

↓

LLM

↓

JSON

↓

Google Form

↓

Link

This is your first complete MVP.

Phase 6 – AI Agent

Instead of a single LLM call, create a multi-step agent.

User Prompt



Planner



Research Node



Question Generator



Validator



Google API Tool



Publisher



Return Links

Each node has a specific responsibility, making the system more reliable and extensible.

Phase 7 – Conversational Editing

Example:

User:

Add a phone number field.

Agent:

Updates the existing form.

Other examples:

- Remove question 3
- Make email optional
- Add another section
- Change the title
- Reorder questions

This requires maintaining the form's state.

Phase 8 – Advanced Features

Examples:

- Conditional logic
 - Quiz mode
 - Answer keys
 - Branching
 - File uploads
 - Themes
 - Confirmation messages
 - Response spreadsheet
 - QR code generation
-

Phase 9 – Smart Form Generation

Examples:

Prompt

Internship Registration

The AI infers:

- Name
- College
- Department
- Resume
- GitHub
- LinkedIn
- Skills
- Preferred domain

No template needed.

Phase 10 – Enterprise Features

- Organization templates
 - Team workspaces
 - Form history
 - Versioning
 - Analytics dashboard
 - Share management
 - API keys
 - Usage limits
 - Billing (if commercialized)
-

Recommended Folder Structure

formgen-ai/

|

|— frontend/

| |— components/

| |— pages/

- | └─ services/
- | └─ hooks/
- | └─ utils/
- |
- └─ backend/
- | └─ api/
- | └─ auth/
- | └─ google/
- | └─ agents/
- | └─ llm/
- | └─ models/
- | └─ schemas/
- | └─ services/
- | └─ tools/
- | └─ utils/
- |
- └─ prompts/
- |
- └─ docs/
- |
- └─ tests/